Sequences and Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Sequence limits**

Determine whether each sequence converges, and find the limit if it does:

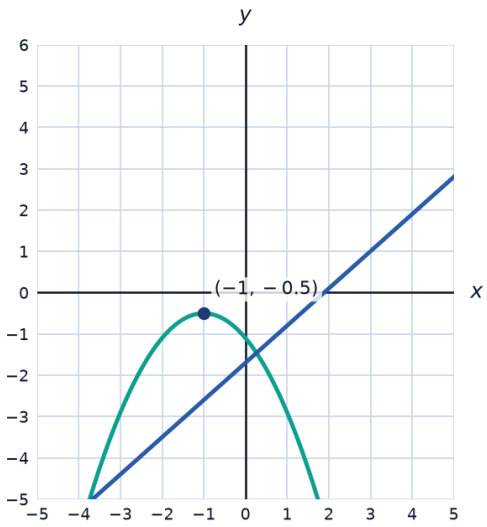
a $a_n = \frac{3n+1}{n+2}$

b $a_n = \frac{n^2}{2^n}$

c $a_n = (-1)^n$

2 **[Foundational]** • **Target: Series and partial sums**

Determine whether $\sum_{n=1}^{\infty} \frac{1}{n}$ converges or diverges, and name the test or fact that justifies your answer.



3 **[Foundational]** • **Target: Series and partial sums**

Evaluate the geometric series $\sum_{n=0}^{\infty} 3\left(\frac{2}{5}\right)^n$.

4 **[Foundational]** • **Target: Series and partial sums**

Use partial fractions and telescoping to evaluate $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

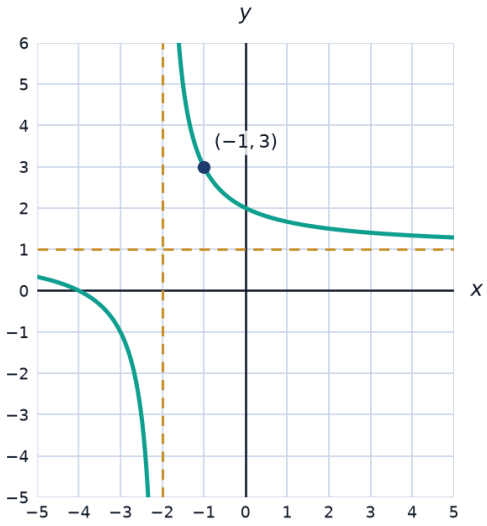
5 **[Foundational]** • **Target: Monotonic and bounded sequences**

Show that the sequence $a_n = \frac{n}{n+1}$ is increasing, by comparing a_{n+1} and a_n.

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6 [Foundational] • Target: Monotonic and bounded sequences

Explain why the sequence $a_n = \frac{n}{n+1}$ (increasing and bounded above by 1) must converge, and find its limit.



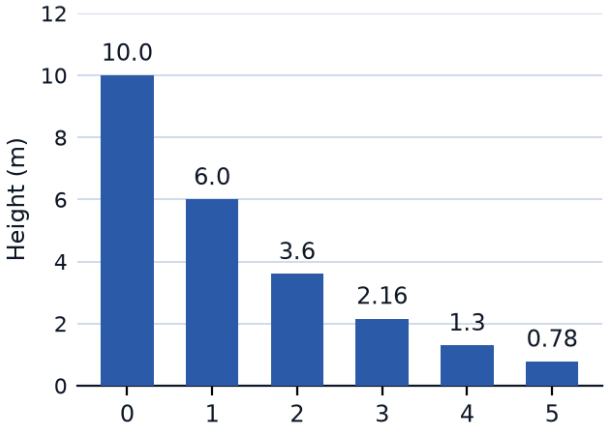
7 [Foundational] • Target: More series practice

Evaluate the series $\sum_{n=1}^{\infty} \left[\left(\frac{1}{2}\right)^n + \left(\frac{1}{3}\right)^n\right]$ by splitting into two geometric series.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: More series practice

A ball dropped from 10 m rebounds to 60% of its previous height each bounce. Using an infinite geometric series, find the total distance traveled.



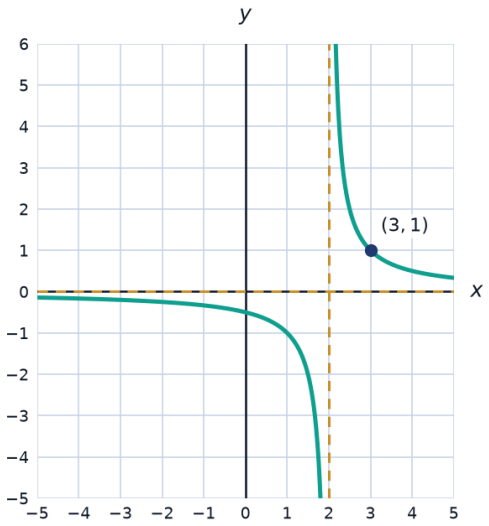
9 [Intermediate] • Target: Recursive sequences

A sequence is defined recursively by $a_1 = 2$, $a_{n+1} = \frac{a_n + 6}{2}$. Compute a_2, a_3, and a_4.

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10 **[Intermediate]** • **Target: Recursive sequences**

Assuming the recursive sequence above converges to a limit L, find L by setting $L = \frac{L+6}{2}$ and solving.



11 **[Intermediate]** • **Target: Repeating decimals as series**

Express the repeating decimal $0.\overline{45}$ as a fraction using an infinite geometric series with $a = \frac{45}{100}$ and $r = \frac{1}{100}$.

12 **[Intermediate]** • **Target: Visualizing partial sums of a series**

The graph shows the first six partial sums $S_n = \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$ of a geometric series. Use the pattern to conjecture the value the sums are approaching, and confirm it using the geometric series formula.

13 **[Intermediate]** • **Target: Series with factorial and combined terms**

Evaluate $\sum_{n=0}^{\infty} \frac{1}{2^n} - \sum_{n=0}^{\infty} \frac{1}{3^n}$ by evaluating each geometric series separately.

14 **[Intermediate]** • **Target: Series with factorial and combined terms**

A sequence is defined by $a_n = \frac{2n + (-1)^n}{n}$. Find $\lim_{n \to \infty} a_n$, explaining why the $(-1)^n$ term does not affect the limit.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

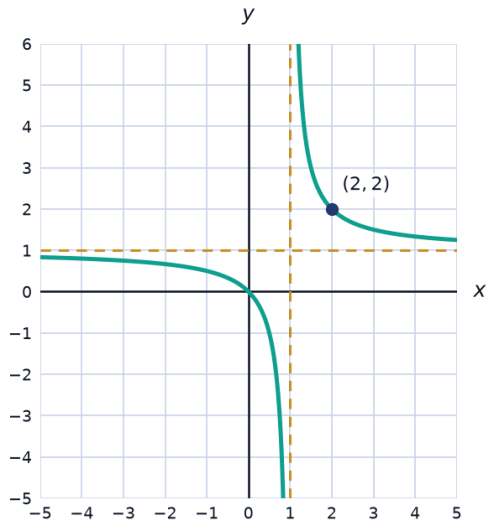
15 **[Challenging]** • **Target: The Squeeze Theorem for sequences** • **Cross-Lesson: 6.02 Optimization and Curve Sketching**

Use the squeeze theorem to find $\lim_{n \to \infty} \frac{\sin n}{n}$, using the fact that $-1 \le \sin n \le 1$ for all n.

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16 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 7.01 Integration by Substitution

Use the squeeze theorem to find $\lim_{n\to\infty} \frac{n!}{n^n}$, using the bound $0 < \frac{n!}{n^n} \le \frac{1}{n}$ for $n \ge 1$.



17 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 7.02 Integration by Parts

A patient takes a 100 mg dose of medication every 8 hours. Between doses, 75% of the drug remaining is eliminated, so a fraction $r = 0.25$ carries over to just before the next dose. Using an infinite geometric series, find the long-run steady-state amount in the body immediately after a dose (assume doses accumulate indefinitely).

18 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 8.01 Area Between Curves

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

19 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 8.02 Volumes of Revolution

[Sequences & Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

20 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

Level 1 · FoundationsVerify that the Mean Value Theorem applies to $f(x) = x + \frac{1}{x}$ on $[1, 4]$, and find all values of c that satisfy the theorem.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Sequence limits

The sequence of partial sums $S_n = 2 - 1/2^n$: this means $\sum a_n =$

a. 1

b. 3/2

c. 2

d. diverges

22 [Foundational] • Target: Series and partial sums

A monotone bounded sequence:

a. diverges

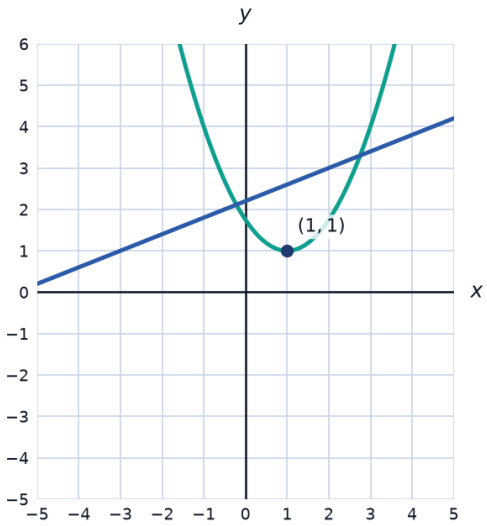
b. converges

c. oscillates

d. is arithmetic

23 [Foundational] • Target: Monotonic and bounded sequences

An increasing sequence bounded above:



a. diverges

b. oscillates

c. converges

d. is arithmetic

24 [Foundational] • Target: More series practice

Squeeze theorem for sequences: if $a_n \le b_n \le c_n$ and $a_n, c_n \to L$, then $b_n \to$

a. L

b. $L/2$

c. 0

d. $2L$

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25 [Foundational] • Target: Recursive sequences

Dirichlet test: $\sum a_n b_n$ converges if a_n monotone $\to 0$ and partial sums of $\sum b_n$ are:

a. convergent

b. zero

c. increasing

d. bounded

26 [Foundational] • Target: Repeating decimals as series

A sequence is bounded above if there exists M such that $a_n \leq M$ for all n. This bound is:

a. always 0

b. the limit

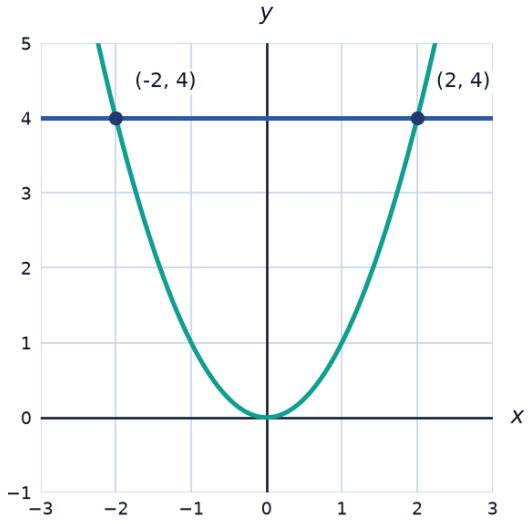
c. not necessarily unique

d. equal to the supremum

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27 [Foundational] • Target: Visualizing partial sums of a series

Consider the region bounded by the curves $y = x^2$ and $y = 4$. This region is revolved around the line $y = 5$ to form a solid. A cylindrical hole is drilled through the center of this solid, perpendicular to the axis of revolution. The radius of the cylindrical hole is r. The volume of the material removed by the cylindrical hole is given by $V_{\text{hole}} = \pi\int_a^b (R(x)^2 - r^2)dx$, where $R(x)$ is the outer radius of the solid at a given x and a, b are the limits of integration. Find the value of a and b, and the expression for V_{hole} in terms of r. The question asks for the setup of the integral representing the volume of the material removed.



a. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((5 - x^2)^2 - r^2)dx$

b. Limits of integration are 0 and 4; $V_{\text{hole}} = \pi\int_{0}^{4}((4 - x^2)^2 - r^2)dx$

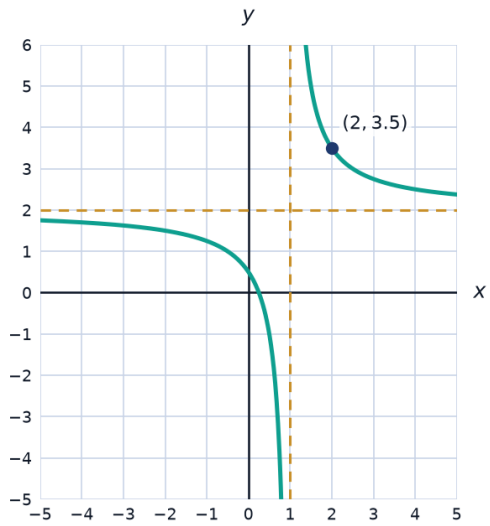
c. Limits of integration are 1 and 4; $V_{\text{hole}} = \pi\int_{1}^{4}((5 - x^2)^2 - r^2)dx$

d. Limits of integration are -2 and 2; $V_{\text{hole}} = \pi\int_{-2}^{2}((4 - x^2)^2 - r^2)dx$

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28 [Foundational] • Target: Series with factorial and combined terms

Consider the sequence defined by $a_1 = 1$ and $a_{n+1} = \frac{a_n^2 + k}{2}$ for $n \geq 1$, where k is a positive constant. If the sequence converges to a limit L, prove that $L^2 - 2L + k = 0$. Furthermore, if the limit L is an integer and $k = 3$, determine the exact value of the limit.



a. L = 3

b. L = 1

c. L = -1

d. $L = \sqrt{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: The Squeeze Theorem for sequences

Consider the region R bounded by the curve $y = \frac{1}{x^2 + 1}$, the x-axis, and the vertical lines $x = 0$ and $x = 1$. A solid is formed by rotating R about the y-axis. Determine the volume of this solid using the method of cylindrical shells. Express your answer in terms of π.

a. $\frac{\pi}{2}$

b. $\pi \ln(2)$

c. π

d. $\frac{\pi}{4}$

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30 [Intermediate] • Target: Applications of infinite series

Consider the arithmetic sequence with the first term $a_1 = 5$ and a common difference $d = 3$. What is the value of the 5th term, a_5?

a. 17

b. 20

c. 15

d. 23

31 [Intermediate] • Target: Sequence limits

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 8

b. 9

c. 7

d. 10000

32 [Intermediate] • Target: Series and partial sums

A sequence $\{a_n\}$ converges if $\lim_{n\to\infty} a_n$ exists and is:

a. finite (a real number)

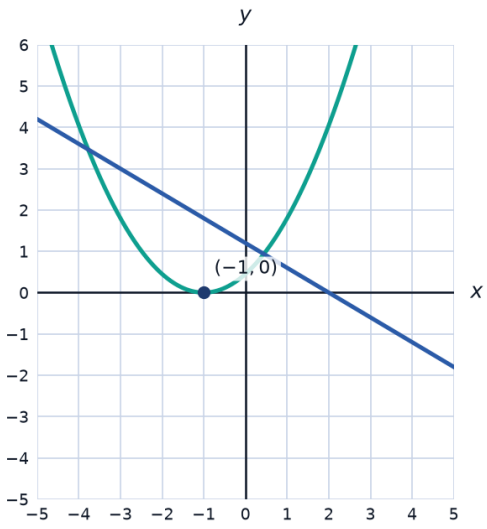
b. infinite

c. 0

d. positive

33 [Intermediate] • Target: Monotonic and bounded sequences

Sum of first n terms of arithmetic series: $S_n =$



a. na_1

b. $a_1 + nd$

c. $n(a_n - a_1)$

d. $\frac{n}{2}(a_1 + a_n)$

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34 [Intermediate] • Target: More series practice

Squeeze theorem: $-x^2 \le f(x) \le x^2$ near $x = 0$ implies $\lim_{x \to 0} f(x) =$

a. 0

b. ± 1

c. DNE

d. 1

35 [Intermediate] • Target: Recursive sequences

Sum of infinite geometric series ($|r| < 1$): $\sum_{k=0}^{\infty} ar^k =$

a. $\frac{a}{r}$

b. $\frac{a}{1-r}$

c. $a(1-r)$

d. $\frac{a}{1+r}$

36 [Intermediate] • Target: Repeating decimals as series

Partial sum $S_n = \frac{2n}{n+1}$. Sum of series:

a. 0

b. 1

c. 3

d. 2

37 [Intermediate] • Target: Visualizing partial sums of a series

Fibonacci sequence: $1, 1, 2, 3, 5, 8, 13, \ldots$. The recurrence is $a_{n+2} =$

a. $a_n + 1$

b. $2a_{n+1}$

c. $a_{n+1} + a_n$

d. $a_{n+1} - a_n$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Series with factorial and combined terms • Cross-Lesson: 8.01 Area Between Curves

Mertens' theorem: if $\sum a_n = A$ (absolutely) and $\sum b_n = B$, then the Cauchy product series $\sum c_n$ converges to:

a. $A + B$

b. $A - B$

c. A/B

d. AB

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39 [Challenging] • Target: The Squeeze Theorem for sequences • Cross-Lesson: 8.02 Volumes of Revolution

Sum of n terms of geometric $1 + 2 + 4 + \cdots$: $S_n =$

a. $2^n - 1$

b. 2^n

c. $n \cdot 2^n$

d. 2^{n+1}

40 [Challenging] • Target: Applications of infinite series • Cross-Lesson: 8.03 Arc Length and Improper Integrals

The sequence $a_n = \frac{n+1}{n}$ converges to:

a. 0

b. ∞

c. 2

d. 1

41 [Challenging] • Target: Sequence limits • Cross-Lesson: 9.01 Separable Differential Equations

Geometric sequence: $a_n = a_1 \cdot r^{n-1}$. If $a_1 = 2, r = 3$, then $a_4 =$

a. 6

b. 18

c. 54

d. 162

42 [Challenging] • Target: Series and partial sums • Cross-Lesson: 1.01 Precalculus Review for Calculus

Arithmetic sequence: common difference d. If $a_1 = 3, d = 4$, then $a_5 =$

a. 15

b. 19

c. 20

d. 23

43 [Challenging] • Target: Monotonic and bounded sequences • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Find $\sum_{n=1}^{\infty} \frac{1}{n(n+2)}$ using partial fractions:

a. 1

b. 1/2

c. 2/3

d. 3/4

44 [Challenging] • Target: More series practice • Cross-Lesson: 12.01 Convergence Tests

Squeeze Theorem: If $-x^2 \le f(x) \le x^2$ near $x = 0$, then $\lim_{x \to 0} f(x) =$

a. -1

b. 0

c. 1

d. cannot determine

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45 [Challenging] • Target: Recursive sequences • Cross-Lesson: 12.02 Power Series and Taylor Series

The sequence $a_n = \frac{\sin n}{n}$: does $\sum a_n$ converge?

a. Yes (by Dirichlet test)

b. No

c. Yes (absolutely)

d. Inconclusive

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